Donald Khek

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Education

University of Florida Gainesville, FL
Bachelor of Science in Electrical Engineering *Expected May 2028*

- **Cumulative GPA**: 3.89/4.0
- **Coursework**: C++ for Engineers, Computer Aided Graphics and Design, Digital Logic and Computer Systems, Signals and Systems

Technical Skills

Languages : Python, JavaScript, HTML/CSS, Lua, C++, LaTeX

Hardware & Tools : KiCad, SolidWorks (CSWA Certified), Git, Docker, SPI, I2C, HiPerGator HPC (SLURM)

Frameworks & Libraries : PyTorch, NumPy, Web Audio API, Chrome Extensions (MV3), REST APIs

Operating Systems : Windows, Linux, Unix/macOS

Experience

TEA Lab Jun. 2026 – Present
Undergraduate Research Assistant | Python, NumPy, HiPerGator HPC *Gainesville, FL*

- Engineered parallelized data generation on the **HiPerGator HPC Cluster** using custom Bash and SLURM scripts, creating images and CSV data with **Python** and highway-env
- Developed and trained a Convolutional Neural Network (CNN) using **PyTorch** and ResNet-18 to predict the time-to-collision from images, achieving a Mean Square Error (MSE) of 0.0035
- Participated in weekly architecture reviews to direct data collection pipelines, evaluate loss convergence, and optimize model hyperparameters

Projects

Gatorlator (Chrome Extension) | GitHub | *Javascript, Chrome API, Web Audio* Jan. 2026

- Awarded **Best Education, Accessibility, & Social Impact Project** at SwampHacks 2026
- Engineered a Chrome extension to capture active tab audio, achieving **sub-500ms latency** for real-time translation and TTS to support accessibility in educational environments
- Integrated **Deepgram** for speech-to-text, **DeepL** for translation, and **ElevenLabs** for high-quality audio output, managing asynchronous API calls and audio streams within a Chrome Service Worker using Javascript

Custom E-Paper E-Reader | *C++, KiCad, ESP32, Hardware Design, SPI, I2C* July 2026 – Aug. 2026

- Designed and routed a compact **4-layer PCB** in **KiCad** around an **ESP32-C3** RISC-V microcontroller, integrating an onboard boost converter circuit to supply high-voltage bias rails for a **5-inch E-Ink** display
- Implemented an automated **P-MOSFET power-path** switching circuit with **MCP73831** LiPo charging management, optimizing battery longevity with hardware power-gating
- Ported and refactored the open-source **CrossPoint** C++ firmware, optimizing memory footprint and partition tables to run reliably on **4MB Flash** (down from 16MB)

Personal Home-Lab & Network Infrastructure | *Debian, Raspberry Pi, Docker* Dec. 2025 – Jan. 2026

- Engineered a dual-node home server environment using a Raspberry Pi 3 B+ and a dedicated Debian-based PC to host network-wide services
- Deployed **Pi-hole** for DNS-level ad blocking and configured **Tailscale** as a secure VPN exit-node, enabling encrypted remote access to the home network
- Implemented **Up-Snap** for Wake-on-LAN (WoL) functionality and utilized **Pterodactyl** to manage containerized game server instances, **improving hardware utilization by 25%** through strict resource capping and isolation